Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The UDP protocol reveals that the browser sent a DNS request to the DNS server using UDP port 53. This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: "UDP port 53 unreachable." The port noted in the error message is used for DNS (Domain Name System) services, which resolve domain names into IP addresses. The most likely issue is that the DNS server was not accepting requests on UDP port 53 because the DNS service was unavailable, misconfigured, or blocked, preventing the domain name from being resolved.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
The incident occurred at 13:24:32.192571, when multiple users reported that they were unable to access www.yummyrecipesforme.com and received the error message "Destination Port Unreachable." The cybersecurity team responded by reproducing the issue and capturing network traffic using the tcpdump protocol analyzer. The packet analysis revealed that DNS requests sent over UDP port 53 were unsuccessful because the DNS server returned ICMP "UDP port 53 unreachable" messages. As a result, the browser could not resolve the website's domain name into an IP address, preventing users from accessing the website. The investigation is ongoing to determine the root cause of the issue. The next steps include verifying that the DNS service is running correctly, checking whether UDP port 53 is blocked by a firewall, reviewing the DNS server configuration, and examining the server logs for signs of misconfiguration or malicious activity. At this stage, the most likely cause is that the DNS service on UDP port 53 is unavailable, misconfigured, or being blocked.